



Assessment/Theory

ACL+NAT-AS-01

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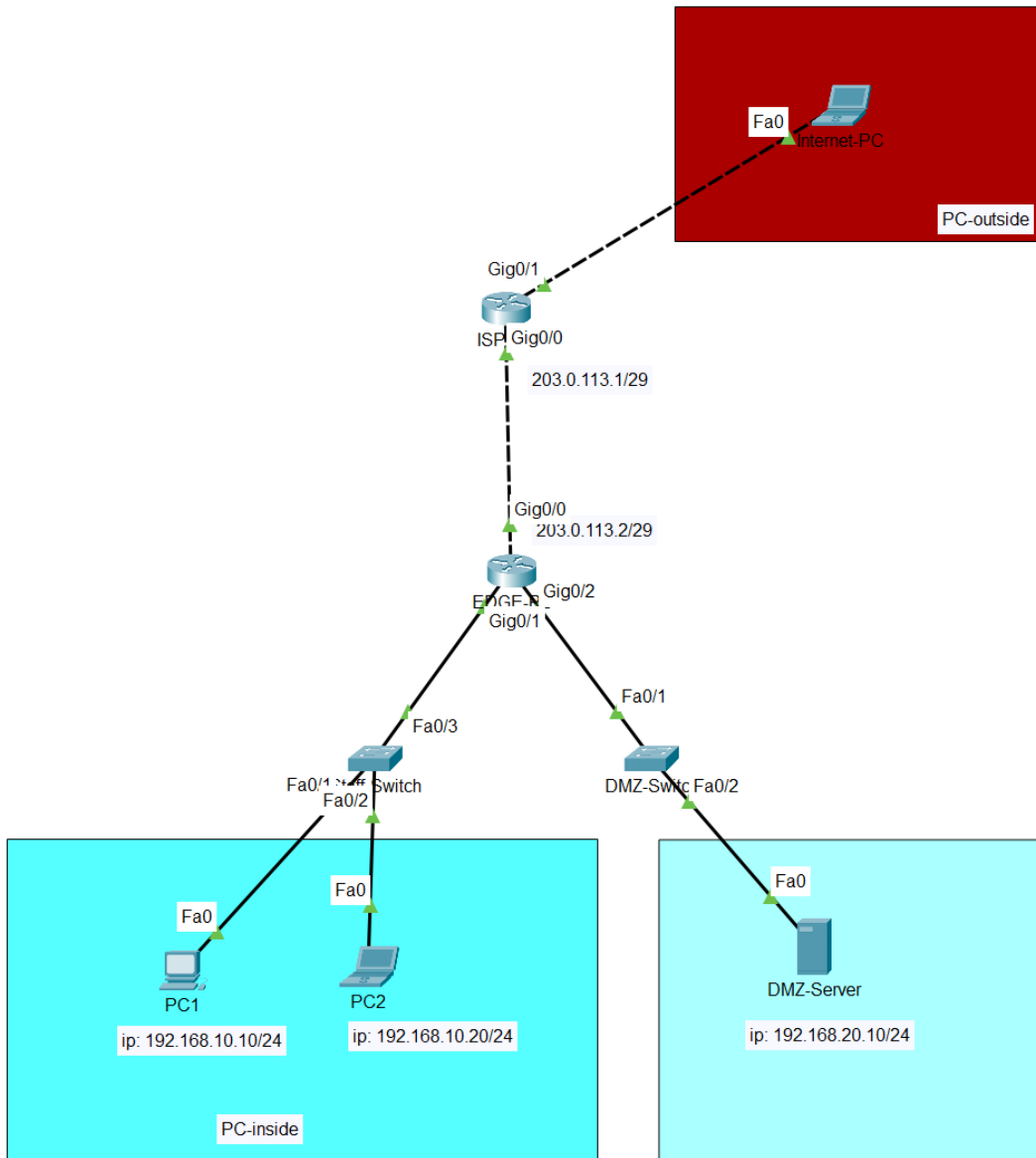
Class: CYG128C3

Instructor: Aun Serey

4. Assessment / Theory Questions for Students

You can use these as written answers or oral questions:

Here is my topology:



A. NAT & Verification

1. Run show ip nat translations on EDGE-R2:

- Identify the PAT entries created when PC1 browses the Internet.

Answer:

```
EDGE-R2#show ip nat translations
Pro Inside global      Inside local      Outside local     Outside global
icmp 203.0.113.2:4      192.168.10.10:4   203.0.113.1:4     203.0.113.1:4
icmp 203.0.113.2:5      192.168.10.10:5   203.0.113.1:5     203.0.113.1:5
icmp 203.0.113.2:6      192.168.10.10:6   203.0.113.1:6     203.0.113.1:6
--- 203.0.113.3         192.168.20.10    ---               ---
EDGE-R2#
```

- Inside Local (192.168.10.10):** identifies the traffic is coming from **PC1**.
- Inside Global (203.0.113.2):** identifies that the private IP was translated to the Router's **Outside Interface IP**.
- The Port Numbers (:4, :5, :6):** This is the key proof of **PAT (Port Address Translation)**.

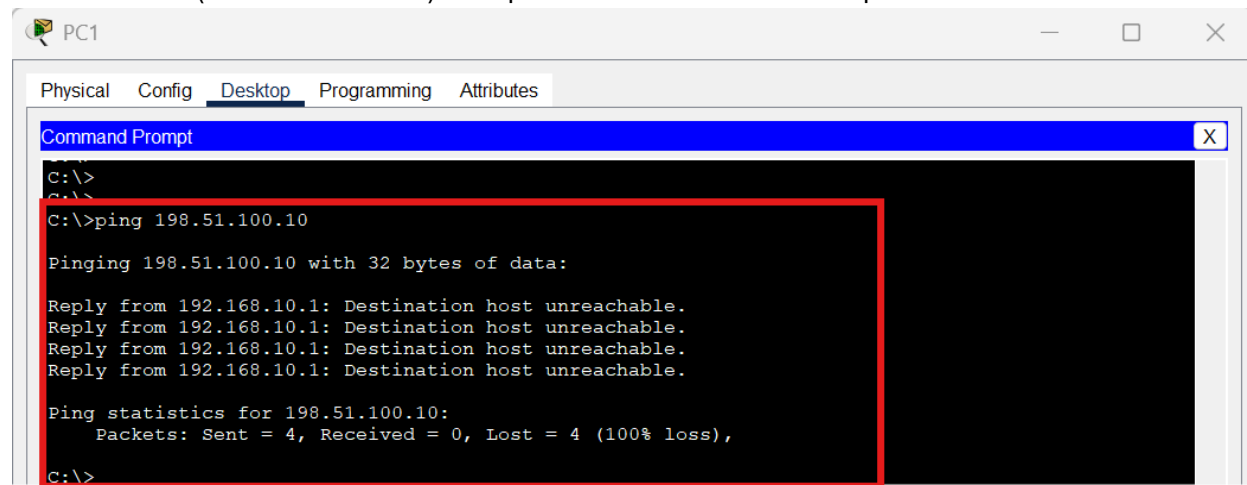
B. Routing vs NAT

1. If you remove this route on EDGE-R2:

ip route 0.0.0.0 0.0.0.0 203.0.113.1

- What happens to Internet access from PC1?

Answer: NO, the connection will fail. After I remove default route I'm trying to ping to Internet-PC (198.51.100.10/24) from pc 1 the result shows as the picture below

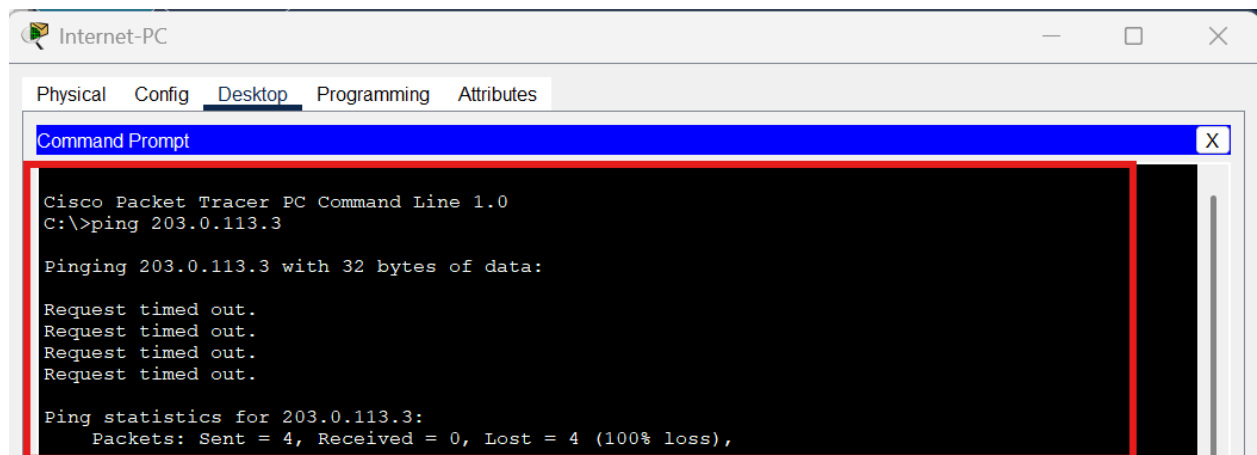


- Can PC-Outside still reach 203.0.113.3? Explain.

Answer: No, the connection will fail

Explain:

- For connection work, for ping successfully we need both direction request and reply, but for this question only request(inbound) work and reply(outbound) fail because I deleted the default route that cause the EDGE-R2 does not know where 192.168.100.10 is.
- One more thing from the previous point of lab requirement routing on ACL policy require that **From Internet → DMZ: Only HTTP (TCP/80) allowed to 203.0.113.3. All other traffic to that IP is denied.** Show it means the connection was blocked and filter by ACL too.



C. ACL Logic & Direction

1. What happens if you apply OUTSIDE_TO_DMZ outbound on G0/0 instead of inbound?

Answer: The security policy **fails completely**, and DMZ not safe and unprotected because we want to filter traffic coming from the internet if we use outbound the traffic will check only packet that leave from the router and go back to the internet so it mean when the packet from internet outside, the router let the packet without filter traffic so it not safe and use more CPU of router to check route before do ACL.